QUANTUM@MTSU · TN-QUMAT · DOE RENEW

AMPLIFY

Scholar Program Su'2026

Paid 10 Weeks Summer Research in Quantum Materials

MTSU · TSU · FISK · MEHARRY · ORNL · U OF MICHIGAN

SUBMIT APPLICATION BY MAY 15TH, 2026

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Are you an undergraduate STEM student curious about quantum science? **AMPLIFY** is a competitive, **paid 10-week summer research program ($3,500; 17.5/hour; 20hrs/week)** that matches you with a faculty mentor to conduct real quantum research, professional development and gives you everything you need to compete for graduate school, NSF fellowships, and a career in the quantum workforce.

This program is funded by a **U.S. Department of Energy RENEW grant** through the **TN-QuMat -ORNL consortium**.

WHAT YOU GET

RESEARCH Faculty-mentored research using HPC supercomputer and experimental research facilities real science.	**PAPER READING** Weekly journal clubs, primary literature workshops, and scientific writing training to build your academic foundation.	**GRAD SCHOOL PREP** Dedicated coaching on Ph.D. applications, personal statements, and connecting with top quantum science programs.
POSTER TRAINING Guided poster design, rehearsal sessions, and presentation at MTSU Scholars Week and national conferences.	**1-ON-1 MENTORSHIP** Weekly mentoring from MTSU, TSU, Fisk, and ORNL scientists plus biweekly professional development workshops.	**FELLOWSHIP SUPPORT** Guided applications for NSF REU, DOE SULI at ORNL, NSF GRFP, and the Goldwater Scholarship.

TN-QUMAT RESEARCH TRACKS

A MODELING AND SIMULATIONS OF QUANTUM SYSTEMS Theoretical & computational study of strongly correlated electrons, disorder effects, and quantum phase transitions using DMFT and quantum Monte Carlo.	**B FIRST-PRINCIPLES SIMULATIONS OF QUANTUM MATERIALS** DFT and ab initio methods applied to topological insulators, Weyl semimetals, and materials with protected surface states for quantum information science.	**C SYNTHESIS & CHARACTERIZATION OF LOW-DIMENSIONAL QUANTUM STRUCTURES; QUANTUM OPTICS** Lab synthesis, thin-film growth, and spectroscopy of 2D materials and novel superconductors using ORNL neutron scattering facilities; quantum optics.

WHO & WHAT COMES NEXT

WHO SHOULD APPLY

→ Undergrad STEM student at MTSU, TSU, Fisk, or Meharry

→ Any year — no prior quantum research required

→ Physics, chemistry, CS, math, biology, engineering, data science

PATHWAYS AFTER AMPLIFY

- ✦ DOE SULI Internship at Oak Ridge National Lab
- ✦ NSF REU, NSF GRFP
- ✦ Goldwater Scholarship, Ph.D programs, URECA
- ✦ Ph.D. programs at top quantum science institutions
- ✦ MTSU URECA undergraduate research grant

Questions? Contact Dr. Hanna Terletska hanna.terletska@mtsu.edu · tn-qumat.org ·